

# Rank of a Matrix: Echelon and Normal Forms

## Understanding Rank Through Inspection

Before applying algorithms, it is often useful to look for linear dependencies. If a row is a exact multiple of another, or if two rows are identical, they are linearly dependent and will eventually become a zero row during reduction.

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## Row Echelon Form for Problem 1

To find the rank, we reduce the matrix to a form where all entries below the leading diagonal (pivots) are zero. The rank is the number of non-zero rows remaining.

### Solving Matrix (i)

$$A = \begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 1 & a & b & 0 \\ 0 & c & d & 1 \end{bmatrix}$$

#### Step 1: Observation

Notice that  $R_1 = R_3$  and  $R_2 = R_4$ . This immediately tells us that two rows are redundant.

#### Step 2: Perform Row Operations

We subtract  $R_1$  from  $R_3$  and  $R_2$  from  $R_4$ :

$$\begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 1 & a & b & 0 \\ 0 & c & d & 1 \end{bmatrix} \xrightarrow{(R_3 \rightarrow R_3 - R_1)} \begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 0 & 0 & 0 & 0 \\ 0 & c & d & 1 \end{bmatrix} \xrightarrow{:, \quad : R_4 \rightarrow R_4 - R_2} \begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

#### Result for (i):

The number of non-zero rows is **2**.

Therefore, **Rank = 2**.

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## Normal Form for Problem 2

Normal form (also called Canonical form) involves using both **Row Operations** and **Column Operations** to transform a matrix into an identity-like block structure:

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Where  $I_r$  is an identity matrix of order  $r$ . The value of  $r$  is the Rank.

## Solving Matrix (ii)

$$A = \begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

### Step 1: Get a '1' at the (1,1) position

Swap  $R_1$  and  $R_3$  to make calculations easier:

$$\begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} (R_1 \leftrightarrow R_3) \longrightarrow \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & -3 & -1 \\ -2 & -1 & -3 & -1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

### Step 2: Clear the first column and first row

( $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 + 2R_1$ ) and Column operations ( $C_3 \rightarrow C_3 - C_1, C_4 \rightarrow C_4 - C_1$ ):

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & -3 & -1 \\ -2 & -1 & -3 & -1 \\ 0 & 1 & 1 & -1 \end{bmatrix} (Row Ops) \longrightarrow \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & -4 & -2 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} (Col Ops) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & -4 & -2 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

### Step 3: Simplify the remaining block

Swap  $R_2 \leftrightarrow R_4$ :

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & -4 & -2 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} (R_2 \leftrightarrow R_4) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & 2 & -4 & -2 \end{bmatrix}$$

### Step 4: Clear entries below and beside the second pivot

Row operations ( $R_3 \rightarrow R_3 + R_2, R_4 \rightarrow R_4 - 2R_2$ ) and Column operations ( $C_3 \rightarrow C_3 - C_2, C_4 \rightarrow C_4 + C_2$ ):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & 2 & -4 & -2 \end{bmatrix} (Row Ops) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix} (Col Ops) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

### Step 5: Final transformation

Swap  $R_3 \leftrightarrow R_4$  and scale  $R_3$  by  $-1/6$ :

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix} (R_3 \leftrightarrow R_4, R_3 \rightarrow -1/6 R_3) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This is the Normal Form  $[I_3 : 0]$ . Since the identity matrix is of order 3:

**Rank = 3.**